



BOTSWANA UNIVERSITY OF AGRICULTURE AND NATURAL RESOURCES

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FRESHWATER MANAGEMENT SYSTEMS

LITERATURE REVIEW

1. Mosepele, B, et al. (2021), talks about how the Okavango Delta's aquatic biodiversity responds to human disturbances like insecticide spraying, cattle encroachment on seasonal floodplains and pollution. The author also indicates the resilience framework to advocate for adaptive management strategies. This study demonstrates the complex interactions between human livelihoods and freshwater ecosystem.
2. Linke et al. (2023), conducted a review about freshwater species from the Okavango delta. It indicates that upstream water and planned hydropower structures and the impacts of climate change threaten the region's rich aquatic biodiversity.
3. The UNESCO World Heritage Centre Okavango Delta Nomination Dossier describes Okavango as an inland covering 2 million hectares in north western Botswana. This study indicates wetlands and the goods and services it provides to surrounding communities and wildlife.
4. A report from population connection and UNU-INWEH (2026) titled "From Crisis to Bankruptcy: The state of Global Freshwater Systems: describes that the water bankruptcy in which more than half of the world's large lakes have shrunk since the early 1990's and approximately 35% of global wetland area has been lost and freshwater vertebrate populations have declined by probably 85% over the past 5 decades.
5. Swatuk and Rahm (2004, updated 2022) conducted policy analysis of Botswana's National Water Master Plan and water management reforms. It indicated that the governance dimensions of freshwater threats and technical solutions alone are not enough without policy implementation and community engagement.
6. Science Direct (2026), 'Conservation Status and Threats to freshwater Forested Wetlands' show that wetlands cover 13% of the Earth's surface. They are described to be highly vulnerable to hydrological alteration, land use change and climate related stressors.
7. Wiley/AGU(2024) projected future climate scenarios and indicated that Central Including Okavango Basin may experience increases in wetland extent. This highlights the importance of protecting existing wetland areas.
8. Springer Nature (22018), 'Wetlands in a changing climate: science, policy and management' states that an increased atmospheric carbon dioxide may nearly double freshwater wetland methane emissions, creating a significant positive feedback loop that speeds up climate change.
9. Wiley/AGU (2024) highlights that the indirect hydrological consequences of land use change and how it would affect agricultural expansion when water availability is less in tropical and subtropical wetland basins.
10. Published review in Frontiers in Conservation Science (2025), identified threats to aquatic biodiversity. It shows that land use change pollution, overharvesting, dam construction and climate change as major threats. It is important to know threats facing Botswana's freshwater ecosystems.

INTRODUCTION

Freshwater systems are aquatic environments containing water with a low concentration of dissolved salts less than 1%. They represent 3% of the Earth's total water. These systems are vital for supporting life, livelihoods and ecosystem. Botswana is classified as a semi-arid country with low and variable rainfall, it has significant freshwater resources like perennial rivers, seasonal streams, pans and wetlands.

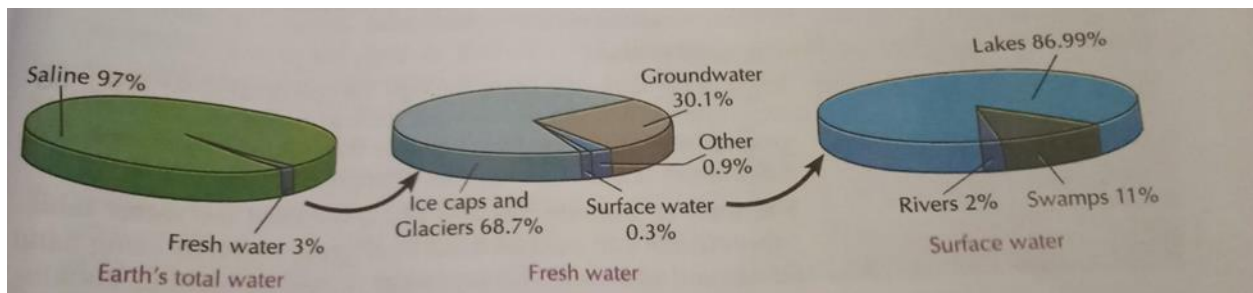


Figure 1. Showing the presentation of freshwater in the Earth.

Wetlands are defined as areas of marsh, fen, and peatland or water whether permanent or temporary with what that is static or flowing fresh or salt including areas of marine water and the depth of which low tides does not exceed six meters. (Mitsch & Gosselink, 2015).

Okavango Delta which is recognized internationally for its ecological importance and high biodiversity. The Delta is formed by the Okavango River, which originates in the highlands of Angola and flows approximately 1000km southward before spreading into Kalahari sands and flows through Namibia before entering Botswana. (Murray-Hudson et al., 2006). The Delta's ecological importance is recognized through its designation as both a Ramsar Site that was founded in 1997 and a UNESCO World Heritage Site which was founded in 2014 placing it among the planet's most significant freshwater systems. (Ramsar Convention Secretariat, 2016).

Despite the importance of freshwater systems in Botswana and across the globe are under a lot of threat. The study site located near the Gaborone Dam consisted of a pond, this pond represents a lentic environment showing minimal water movement.

This report examines threats to freshwater and wetlands systems at a global, regional and local scale, focusing on three key threats: water pollution, habitat degradation and impacts of climate change and altered hydrology.

OBJECTIVES

- To review peer reviewed literature on threats to freshwater and wetland systems at global, regional (African) and local (Botswana) scales.
- To identify and analyze three primary threats to Botswana's freshwater and wetland systems

- To evaluate the interactions between biotic and abiotic factors, land use and freshwater ecosystem health in the study area.
- To propose evident based management and conservation strategies for Botswana's freshwater systems.

MATERIALS AND METHODS

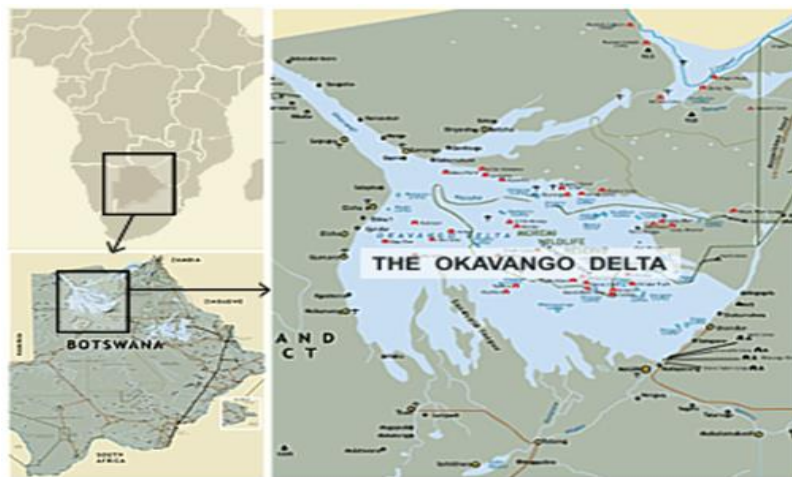
STUDY AREAS DESCRIPTION

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1. Okavango Delta
2. Pond (In the vicinity of Gaborone Dam)

OKAVANGO DELTA

The coordinates for the Okavango Delta are 19°19'7.32"S and 22°26'48.90"E and Delta is characterized by a mosaic of permanent swamps, seasonal floodplains and lagoons. It covers approximately 22 000 km² during peak flooding. The Delta supports high levels of biodiversity making it one of the most biologically productive ecosystems in Africa. The delta covers is important to local communities supporting fishing, tourism and livelihoods.



POND IN THE VICINITY OF GABORONE DAM

The study was conducted in the vicinity of Gaborone Dam, located at coordinates 24°42'0.58" S, 25°55'35" E in the southern district of Botswana. The specific study was a pond situated near the dam, the pond represents a lentic water body which is characterized by slow movement.



The study area is also characterized by shallow turbid water with visible sediment at the bottom, showing limited flow and sediment accumulation. The pond margins are dominated by dense emergent vegetation like reeds and grasses.

METHODOLOGY

OKAVANGO DELTA

The data sourcing about the Okavango Delta was gathered through a literature review of published journals and academic journals like *Invertebrates in Freshwater Wetlands* as well as published reports from the Ramsar Convention to obtain information about biodiversity, ecosystem services and ecological dynamics.

POND IN THE VICINITY OF GABORONE DAM

The Hanna HI98194 multi-parameter was used to measure temperature (°C), Dissolved oxygen (mg/L) and pH. The probe was inserted at a certain depth to ensure accuracy. Then a hand net was used for collecting the samples; an individual was timed 2 minutes to vigorously disturb the littoral zone, ensuring the collection of sediment and small aquatic insects hiding within the water. Once the 2 minutes were completed, the captured material was transferred into a clean container.

The hand net was cleaned to remove any small insects that might be hiding in the net. The samples were then taken to the laboratory to identify aquatic insects such as beetles and other macroinvertebrates were separated for identification. At the laboratory, the water samples were poured into a tray for identification, then were put in a sample container with a few drops of ethanol and were put in the refrigerator. Later, a compound microscope was used to identify an insect and use the “Identification guide to Freshwater Macroinvertebrates” document.



RESULTS

PARAMETER	RECORDED VALUE	THRESHOLD VALUE
pH	7.79	6.5-8.5
Temperature(°C)	25.06 °C	<30 °C
Dissolved Oxygen (mg/L)	5.72 mg/L	>5mg/L

Temperature- Water temperature is important in freshwater ecosystems as it influences oxygen levels and biological processes. The temperatures between 20°C and 30°C are suitable for aquatic organisms. The temperature of 25.06°C indicates the pond has a suitable temperature to support aquatic organisms.

Dissolved Oxygen- Dissolved Oxygen concentration of about 5mg/L is considered the minimum threshold required to support aquatic life in freshwater ponds. Dissolved Oxygen is important for the survival of fish and other aquatic organisms. The recorded value of 5.72 mg/L shows that the pond contains sufficient oxygen to support aquatic life.

pH- The pH of 7.79 shows a slightly alkaline and remains within the acceptable range for freshwater organisms. Extreme pH levels can negatively affect aquatic organisms by physiological stress. High acidic or high alkaline water can damage the gills of fish, interfere with reproduction, and reduce the survival of sensitive species such as fish larvae and aquatic insects.

MACROINVERTEBRATES IDENTIFICATION

	ORDER	FAMILY	COMMON NAME	KEY FEATURES	NUMBER IDENTIFIED
1	Hemiptera	Notonectidae	Backswimmer	• 6 jointed legs • Visible wings • Hinged mouthparts • No tail filaments	4
2	Coleoptera	Scarabaeidae	Beetle larva	• 6 jointed legs • No wing pads • No portable case • No hooks • No prolegs	3
3	Megaloptera	Corydalidae	Dobsonfly larva	• 6 jointed legs • Hooks at the end of the body • Lateral filaments • No portable case	3
4	Odonata	Libellulidae	Dragonfly nymph	• 6 jointed legs • 3 tails filaments • No hooks • No wing pads	5

DISCUSSION

MAJOR THREATS TO BOTSWANA'S FRESHWATER SYSTEM

WATER POLLUTION

The study pond situated in the vicinity of Gaborone Dam is vulnerable to non-point pollution from the surrounding area. The slightly elevated nutrient at the pond is reflected in the borderline dissolved oxygen reading of 5.72mg/L recorded during the field visit which shows that some organic decomposition and possible eutrophication pressure and is a risk that is shown in small, shallow lentic systems.

Globally, pollution is one of the most significant threats to internationally important wetlands, affecting 54% of Ramsar sites worldwide (Hu et al., 2019). In Botswana agricultural pollution threatens wetland systems such as the Okavango Delta, where insecticide spraying for tsetse fly control has historically impacted aquatic invertebrate communities (Mosepele et al., 2021).

To manage water pollution, integrated catchment management approaches are essential like regular water quality monitoring programme should be institutionalized to detect deterioration early and guide adaptive management.

HABITAT DEGRADATION THROUGH LAND USE CHANGE

Land use change is a fundamental driver of freshwater ecosystem degradation in Botswana. In the Okavango Delta region, cattle grazing on seasonal floodplains compacts soils, reduces vegetation cover, and disrupts hydrological connectivity (Mosepele et al., 2021). Across Africa, the conversion of wetlands to farmland has been identified as a leading cause of wetland loss, hence reducing both biodiversity and the ecosystem services these systems provide.

In the study site, the land use surrounding the pond is clearly reflected in the macroinvertebrate community structure. Some of the residential infrastructures observed during the field visit contribute to the habitat degradation as some people are fishing and the area can be used as a grazing site.

Managing habitat degradation requires enforcement of environmental impact assessments for development projects in the catchment area and promotion of sustainable use of land use practices in those areas.

CLIMATE CHANGE AND ALTERED HYDROLOGY

Botswana as a semi-arid country, it is vulnerable to changes in precipitation patterns, increased temperatures and extended drought periods. The Okavango Deltas flooding regime is highly sensitive to changes. Reductions in annual flood pulses after the distribution of habitats affect fishes and reduce the extent of productive wetland areas (Mosepele et al., 2021).

For the study site, shallow water bodies can dry out completely during dry periods, eliminating aquatic communities. Including partial reduction in water level concentrates pollutants and nutrients, raises water temperatures and reduces dissolved oxygen.

Small shallow ponds are especially vulnerable to thermal stress as they heat up faster and have less thermal buffering capacity than large deep reservoirs. The management response to climate change threats must include water demand management, investment in alternative water sources such as wastewater recycling and aquifer recharge, and transboundary cooperation with upstream countries to protect shared river systems.

INTERACTIONS BETWEEN BIOTIC AND ABIOTIC FACTORS

Abiotic factors are non-living components (water chemistry, temperature, light availability, sediment) on the environment while biotic factors are living components on the environment (plants, invertebrates, fish, birds).

High nutrient levels can stimulate algal growth which in turn affects oxygen availability for fish and other organisms. Extreme pH levels can harm macroinvertebrates and other organisms

affecting their reproduction and survival. Also, sunlight helps aquatic plants and algae to photosynthesize hence producing oxygen that help to support other organisms while water flow moves sediments and nutrients which then influences which plants and organisms can survive in different parts of the system.

Such interactions help maintain ecological balance and support life. These interactions are important for freshwater systems in order to ensure that the ecosystem is functioning properly and provide needed services.

Dissolved oxygen, temperature, and pH collectively determine the suitability of a water body for different aquatic organisms. The dissolved oxygen level of 5.72 mg/L recorded at the site sits just above the critical threshold indicating that any further reduction due to increased organic loading, elevated temperatures or reduced water circulation could cause the loss of some macroinvertebrates.

Wetlands and pond (study site) sediments store significant quantities of organic carbon and when these systems are degraded, stored carbon is released as carbon dioxide and methane contributing to climate change.

CONCLUSION

This study identified water pollution, land use change and habitat degradation, and climate change and altered hydrology as the three primary threats to these systems. The study site revealed a moderate quality lentic water body with water quality parameters within the threshold.

The Okavango Delta represents one of the most important wetland ecosystems in the world and supports a wide variety of plant and animal species and providing important services such as water supply, fisheries, tourism and habitat for wildlife.

The Okavango Deltas Ramsar and World Heritage status provides a framework for international cooperation and funding while national policies must be threatened to protect smaller water bodies as they are of much importance.

The pond (study site) is a smaller lentic system but still provides important ecological functions by supporting aquatic plants, insects and small fish while also serving as a water source for nearby communities.

The study also highlights the importance of interactions between abiotic factors such as water quality, temperature and nutrient availability and biotic components including plants, insects, and fish. The interactions regulate ecosystem processes and help maintain the stability and productivity of freshwater habitats. It is important to adopt approaches to freshwater management that account for the complex interactions between human activities, land use, biodiversity and climate.

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